

Aditya Goyal

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EDUCATION

University of Wisconsin-Madison

Bachelor of Science in Computer Science and Data Science

GPA: 3.93/4.0

Madison, WI

Sep. 2023 – May 2027

Relevant Coursework: High Performance Computing, Big Data Systems, ML Research, AI Accelerated Honors in Algorithms, Data Structures, Matrix Methods in ML, Modeling in R, Computer Systems, Database Management

TECHNICAL SKILLS

Languages: Java, Python, C, C++, R, SQL (MySQL, PostgreSQL)

ML: CUDA, PyTorch, scikit-learn, GGML, pandas, NumPy, Matplotlib

Backend/Data Engineering: gRPC, Flask, FastApi, Redis, Spark

Developer Tools: Git, GitHub, Docker, Docker Compose, Postman, Google Cloud, GitHub Copilot, Microsoft Copilot

EXPERIENCE

ML Research Engineering Intern(Yin Li's Lab) | UW Madison Computer Sciences

November 2025 – Present

- Built reproducible mortality forecasting pipeline for 6K sparse, irregularly sampled NICU recordings with missingness and temporal gaps.
- Mitigated label imbalance through focal loss, prevalence-based bias initialization, and positive-aware batch sampling
- Outperformed baselines at high recall, **improving average k fold precision@80% recall from 0.10 to 0.23.**

ML Research Engineering Intern(Sprocket Lab) | UW Madison Computer Sciences

July 2025 – May 2026

- Developed an MIA evaluation pipeline using PyTorch** to test algorithms for reverse-engineering fine-tuning dataset distribution from base and fine-tuned model weights.
- Designed new algorithms, with a benchmarking harness demonstrating **10%** performance gain over baselines. (**accepted to ICML'26 workshop**)

Software Engineering Intern | Frizzle (YC S25)

May 2026 - Present

- Proposed and shipped an AI rubric-generation feature at Frizzle AI using LangGraph and LangSmith, achieving 87% eval accuracy at 0.70\$ per rubric and built scheduled ingestion and eval pipelines for automated RAG ingestion and rubric-quality tracking.

Machine Learning Intern | Freecharge

May 2025 – August 2025

- Improved spend analysis accuracy by 40%** by engineering AI agents with **AWS Bedrock** to orchestrate tools, generate SQL with RAG context, and deliver charts alongside explanations.
- Boosted semantic match accuracy by 30%** by building a search engine using **AWS Kendra** over 12,000+ pages and a preprocessing pipeline isolating high-signal fields from noisy webscraped text.
- Enabled real-time branch analytics with YOLO-based video POCs for entrant counting and opening time detection.

Software Engineering Intern | UW Madison Department of Physics

Nov 2024 – Feb 2025

- Implemented Python scripts to train MLPs on mathematical functions and visualize activation flow across layers.
- Automated large-scale experimentation** by batch processing MLP training runs on randomly generated functions.
- Improved runtime by 25%** and reduced intermediate memory with vectorized operations and broadcasting.
- Built customizable MLP architectures by parameterizing layers and units, enabling user-specified network designs.

Machine Learning Teaching Assistant | UW Madison Computer Sciences

Sept 2024 – Present

- Leading recruitment of future peer mentors and supporting 100+ students every week in CS540 (Intro to AI) and CS 300 (Programming II, Java) on debugging projects and course content, **earning 30+ positive reviews.**

PROJECTS

LLM finetuning using LORA | Python, Pytorch

May 2025

- Fine-tuned FLAN-T5 Large using LoRA for sentiment classification, **reducing trainable parameters by 95%**, achieving 95% accuracy on test set

ShortNExact | Python, FastAPI, Docker, OpenAI SDK, Redis, PgBouncer, HAProxy, Locust

July 2025

- Built a production-grade agentic AI web app that rewrites text to exact word counts while preserving meaning, garnering **2000+ users since deployment.**
- Containerized services with Docker Compose, improving reliability through automated health checks and restart policies.
- Sustained 30+ concurrent users at sub-100 ms median latency** by using **async FastAPI components** and libraries, and validated scalability by load-testing real traffic patterns using **Locust.**
- Reduced latency by 30% and enabled zero-downtime failover** by deploying **PgBouncer** and **HAProxy**, with secure API access enforced via **Redis**, and API key validation via **PostgreSQL DB** and **SQLAlchemy.**

CERTIFICATIONS

• Intro to Cloud Computing (IBM) •DL Specialization (Stanford) •ML Specialization (Stanford) •Mathematics for ML (Stanford)

HOBBIES

• Scuba-diving (PADI certified OPEN Water Diver) • Guitar